

Arsitha Sathu

Software Engineer

Edison, NJ | (202) 222-5321 | sathuarsitha@gmail.com | [LinkedIn](#)

Summary

AI-focused Software Engineer with experience building full-stack and LLM-powered applications using Python, Java, Spring Boot, FastAPI, React, and Angular. Skilled in designing and deploying RAG pipelines, agentic workflows, and multi-agent systems integrated with OpenAI and Azure AI services, alongside scalable REST APIs and microservices on AWS and Azure.

Skills

Programming Languages: Java, Python, C, JavaScript (ES6+), TypeScript, SQL	Database: PostgreSQL, MySQL
Backend Development: Spring Boot, Fast API, Node.js, RESTful APIs, Microservices Architecture	AI/ML & Data Engineering: Machine Learning Basics, LLM Integration, Retrieval-Augmented Generation (RAG), Vector Embeddings, AI-Assisted Development
Frontend Development: React.js, Angular, HTML5, CSS3	Development Tools: Git, Postman, Swagger, JUnit, JIRA, SonarQube, VS Code, IntelliJ IDEA
Cloud Technologies: AWS (EC2, S3, RDS, Lambda, EKS), Azure (AI Foundry, Speech Services, Container Apps, OpenAI (GPT-4o), Azure Key Vault)	Software Engineering Practices: Data Structures & Algorithms, OOP, Agile & Scrum
Containerization & Orchestration: Docker, Kubernetes	

Experience

BLUE HORSE DIGITAL Software Engineer	July 2025 -
- Architected a low-latency, voice agent using Python and Azure AI Services (Speech/GPT-4o), optimizing client-side audio with a custom JavaScript Audio Worklet and implementing barge-in interruption for sub-second, natural responsiveness. - Built a RAG pipeline using Azure OpenAI and an MCP-based agentic workflow to answer questions using USDA policy documents, returning sourced, citation-backed responses. - Engineered a hands-free PDF Form Filling system that dynamically inspects schemas (using PyPDF) to map conversational data into applications in real-time, auto-generating and emailing completed forms without manual input. - Deployed the scalable, containerized application to Azure Container Apps using Docker multi-stage builds, ensuring secure API key management and robust production infrastructure.	
INFOSYS LIMITED Software Engineer	Oct 2021 - July 2023
- Developed production REST APIs and microservices using Spring Boot, Angular, TypeScript, and PostgreSQL for a healthcare compliance application and features that support internal systems. - Optimized database queries and API endpoints, improving response times by 25–30% through strategic indexing and query refactoring. - Built a reusable Angular component library (dynamic tables, filters, role-based access controls) adopted across development teams, reducing UI development time by 30%. - Enforced code quality standards using SonarQube and resolved high-priority production bottlenecks with 90%+ test coverage using JUnit. - Tracked sprints and issues in JIRA while driving continuous integration and code quality within an Agile Scrum methodology.	

Projects

Threat Ledger Next.js, AWS Aurora PostgreSQL, Prisma, OpenAI API
- Built parsers for Suricata, Zeek, and firewall log formats to normalize 200K+ raw alerts into a unified Aurora PostgreSQL schema. - Designed a percentile-based correlation engine that reduced 200+ raw alerts into under 20 scored attack campaigns mapped to kill-chain stages. - Integrated the OpenAI API to auto-generate threat summaries from correlated alert and IP reputation data, cutting manual triage to a single click. - Deployed the scalable, containerized application to Azure Container Apps using Docker multi-stage builds, ensuring secure API key management and robust production infrastructure. - Deployed a production-ready app on Vercel backed by Aurora PostgreSQL with 21K+ IP reputation records.
Clip Learn Python, FastAPI, OpenAI API (GPT-4o), ChromaDB, yt-dlp, JavaScript, HTML/CSS, SSE
- Built a multi-agent AI app (Fast API, Python) that transforms YouTube lectures into study materials, orchestrating 5 specialized agents (transcript extraction, content generation, semantic search, translation, pedagogical audit) via an SSE-based pipeline. - Implemented semantic Q&A using Chroma DB vector search and OpenAI GPT-4o, enabling timestamp-accurate question answering and AI-generated outlines, summaries, and flashcards from raw video transcripts. - Designed a faculty-mode audit feature that evaluates lecture quality across pedagogy, accessibility, and clarity dimensions, and added multi-language support (10 languages) via an on-demand translation pipeline.
Abstractive Text Summarization Python, mT5 Transformer, Git, PyTorch, Transformers, Wandb, Pandas, NumPy
Fine-tuned an mT5 Transformer on a custom dataset of 200+ Telugu narratives, implementing a specialized NLP preprocessing pipeline and achieving optimized performance tracked via ROUGE metrics and Wandb.

Education

Master Of Computer Science George Mason University	2023 - 2025
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Certifications

AWS Certified Cloud Practitioner	Feb 2026
AWS Certified Solutions Architect - Associate	May 2026